Prerequisites — Part 1 of ?

Comprehensive Guide to 3D Rotation Matrices, Determinants, and Euler Angles

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Total time spent: 4 hours and 40 minutes

This document compiles the mathematical and physical foundations of 3D rotation matrices, orthogonality, determinants, elemental rotations, and Euler angles.

1 The Mathematical Foundation: $R^T R = I$

The equation $R^T R = I$, where R^T is the transpose of the rotation matrix R and I is the identity matrix, carries three important consequences for rotation matrices.

1.1 Derivation

Write R in terms of its columns c_1, c_2, c_3, so $R = \begin{pmatrix} c_1 & c_2 & c_3 \end{pmatrix}$. The rows of R^T are the columns of R, so the (i, j) entry of $R^T R$ is the dot product of column i with column j:

$$(R^T R)_{ij} = c_i \cdot c_j$$

Requiring $R^T R = I$ therefore means

$$c_i \cdot c_j = \delta_{ij} = \begin{cases} 1 & i = j \\ 0 & i \neq j \end{cases}$$

which says the columns are unit length and mutually perpendicular (our case) $c_i \cdot c_j = 0 \rightarrow i \neq j$.

Note: Why Parallel Columns are Prohibited
In a pure rigid-body rotation, the column vectors of the rotation matrix represent the transformed orthonormal axes. If any two columns were parallel, it would imply that independent spatial axes have collapsed into the same direction. This destroys the volume and handedness of the coordinate system, causing the determinant to drop to $\det(R) = 0$, which is strictly **prohibited in pure rotation** (since pure rotations require $\det(R) = +1$).

1.2 The Inverse is Just the Transpose ($R^{-1} = R^T$)

Multiplying $R^T R = I$ on the right by R^{-1} gives

$$R^T = R^{-1}$$

– **Meaning:** To **undo** a rotation and return an object to its original state, heavy matrix inverse calculations are unnecessary. You simply **transpose** the matrix by swapping its rows and columns. This drastically accelerates computations in game engines, physics simulations, and spacecraft navigation.

$$R \cdot R_x^T = I$$

$$Rotation \cdot Exact\ Reverse\ Rotation = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = 1$$

– $\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$**:** indicates that the x-axis remains in its place (1 along x) without deflecting to either side.

– $\begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$**:** indicates that the y-axis remains completely in its original position.

– $\begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$**:** indicates that the z-axis remains completely in its original position.

1.3 Preservation of Vector Lengths (Rigid Motion)

When a vector v is multiplied by R, it spins in space without stretching, shrinking, or distorting. The magnitude is unchanged:

$$\|Rv\|^2 = (Rv)^T(Rv) = v^T R^T \cdot Rv = v^T v = \|v\|^2 \quad \Longrightarrow \quad \|Rv\| = \|v\|$$

– **Physical Meaning:** A pure rotation is a rigid-body motion; it changes the spatial orientation of an object but preserves its size, shape, and vector lengths entirely.

– **Step 1 (Squared Norm):** The squared magnitude of the transformed vector Rv is written as the matrix product of the vector's transpose with itself: $(Rv)^T(Rv)$.

– **Step 2 (Transpose Distribution):** Using matrix transposition rules, the transpose of a product reverses the order: $(Rv)^T = v^T R^T$.

– **Step 3 (Rearrangement):** Substituting this back into the expression yields $v^T R^T Rv$.

– **Step 4 (Applying Orthogonality):** Because a rotation matrix satisfies $R^T R = I$, we replace the middle term with the identity matrix: $v^T(I)v$.

– **Step 5 (Identity Reduction):** Since the identity matrix leaves vectors unchanged ($Iv = v$), the expression reduces to $v^T v = \|v\|^2$, proving that $\|Rv\| = \|v\|$.

In short, The length of the vector after the rotation is equal to the length before the rotation

$$\|Rv\| = \|v\|$$

1.4 Orthonormal Columns and Rows

The three column vectors, which give the new directions of the x, y, and z axes after rotation, are:

– **Orthogonal:** perfectly perpendicular to each other (90° angles).

– **Normalized:** each has length exactly 1.

Because $R \cdot R^T = I$ also holds (since $R^T = R^{-1}$ commutes with R), the rows are orthonormal too.

2 The Determinant Condition: $\det(R) = +1$

2.1 What is a Determinant Geometrically?

The determinant det of a matrix is the **scaling factor of volume (or area)** under the transformation:

– $\det = 2$ the object has doubled in volume.

– $\det = 0$ the object has collapsed into a line or point.

– **Rotations** the determinant is restricted to $+1$ or -1.

To see why, take the determinant of $R^T R = I$. Since $\det(R^T) = \det(R)$ and $\det(AB) = \det(A)\det(B)$,

$$\det(R)^2 = \det(I) = 1 \quad \Longrightarrow \quad \det(R) = \pm 1$$

2.2 Why $\det(R) = +1$?

1 **Preservation of volume:** because the determinant is exactly 1, the object's volume is unchanged ($1 \times$ Original Volume). It rotates without warping or scaling.

2 **Preservation of handedness (orientation):** in a 3D coordinate system (x, y, z), the right-hand rule governs axis orientation. A positive determinant ensures that spatial orientation is maintained.

2.3 What if $\det(R) = -1$?

A determinant of -1 represents a **reflection** (like standing in front of a mirror).

Example: looking at a right-handed glove in a mirror makes it look like a left-handed glove. In the physical world, a right-handed glove cannot be turned into a left-handed glove purely by rotating it in space. Thus $\det(R) = -1$ is excluded from pure rigid rotations.

3 How to Calculate a Determinant

3.1 The 3×3 Formula

For a general 3×3 matrix A with entries a_{ij} (row i, column j), expand along the first row:

$$\det A = a_{11}\,(a_{22}a_{33} - a_{23}a_{32}) - a_{12}\,(a_{21}a_{33} - a_{23}a_{31}) + a_{13}\,(a_{21}a_{32} - a_{22}a_{31})$$

- **Signs:** the signs alternate $+, -, +$ across the first row. Each term is an entry times the determinant of the 2×2 matrix left after deleting its row and column.
- **Any row or column:** the result is the same whichever row or column you expand along. Pick the one with the most zeros, since zero entries drop out.

3.2 Worked Examples

For $R_x(\theta)$, expanding along the first row (the 1 is the only nonzero entry in that row except for the zeros):

$$\det(R_x) = 1 \cdot \big(\cos\theta \cdot \cos\theta - (-\sin\theta)(\sin\theta)\big) = \cos^2\theta + \sin^2\theta = 1$$

For $R_y(\theta)$, expanding along the middle row, which is $(0, 1, 0)$, only the entry 1 survives:

$$\det(R_y) = 1 \cdot \det \begin{pmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{pmatrix} = \cos^2\theta + \sin^2\theta = 1$$

For $R_z(\theta)$, expanding along the bottom row, which is $(0, 0, 1)$:

$$\det(R_z) = 1 \cdot \det \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix} = \cos^2\theta + \sin^2\theta = 1$$

3.3 General Rotations via Eigenvalues

For a general rotation R in 3D, the eigenvalues $\lambda_1, \lambda_2, \lambda_3$ satisfy $\det(R) = \lambda_1\lambda_2\lambda_3$. Because R is orthogonal, every eigenvalue has modulus 1. The characteristic polynomial is a real cubic, so it has a real root, and that root is ± 1. The other two eigenvalues are either both real or a complex-conjugate pair $\lambda, \bar{\lambda}$ with $|\lambda| = 1$, so their product is $\lambda\bar{\lambda} = |\lambda|^2 = 1$. Therefore

$$\det(R) = (\pm 1) \cdot 1 = \pm 1$$

and the sign is $+1$ exactly when the real eigenvalue is $+1$. A rotation always has an axis that is left fixed (the eigenvector for eigenvalue $+1$), which is why rotations have $\det(R) = +1$.

4 Anatomy of Elemental Rotation Matrices (R_x, R_y, R_z)

Each elemental matrix rotates every vector about one coordinate axis by an angle θ. The direction is counterclockwise when you look back at the origin from the positive end of the axis (the right-hand rule). Applying the matrix to a point $(x, y, z)^T$ shows exactly what moves:

– $R_x(\theta)$**:** rotates about the x-axis. The x-coordinate is unchanged, and y and z mix:

$$R_x(\theta)\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} \mathbf{1} & \mathbf{0} & \mathbf{0} \\ 0 & \cos\theta & -\sin\theta \\ 0 & \sin\theta & \cos\theta \end{pmatrix}\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} x \\ y\cos\theta - z\sin\theta \\ y\sin\theta + z\cos\theta \end{pmatrix}$$

– $R_y(\theta)$**:** rotates about the y-axis. The y-coordinate is unchanged, and x and z mix:

$$R_y(\theta)\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} \cos\theta & 0 & \sin\theta \\ \mathbf{0} & \mathbf{1} & \mathbf{0} \\ -\sin\theta & 0 & \cos\theta \end{pmatrix}\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} x\cos\theta + z\sin\theta \\ y \\ -x\sin\theta + z\cos\theta \end{pmatrix}$$

– $R_z(\theta)$**:** rotates about the z-axis. The z-coordinate is unchanged, and x and y mix:

$$R_z(\theta)\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ \mathbf{0} & \mathbf{0} & \mathbf{1} \end{pmatrix}\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} x\cos\theta - y\sin\theta \\ x\sin\theta + y\cos\theta \\ z \end{pmatrix}$$

4.1 A Concrete Check

Take $\theta = 90^\circ$, so $\cos\theta = 0$ and $\sin\theta = 1$. Then $R_z(90^\circ)$ sends the unit vector along x to the unit vector along y:

$$R_z(90^\circ)\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$$

This matches the picture: turning the x-axis a quarter turn about z brings it onto the y-axis.

4.2 The Pivot (Anchor) of Each Matrix

– R_x**:** the 1 in the top-left. Points on the x-axis do not move ($x' = x$). The 2×2 block acts in the yz-plane.

– R_y**:** the 1 in the middle. The y-axis is the static pivot ($y' = y$). The x and z coordinates rotate around it.

– R_z**:** the 1 in the bottom-right. The z-axis is the static pivot ($z' = z$). The x and y coordinates rotate in the xy-plane.

The sine terms in R_y sit in the opposite pattern to R_x and R_z because the axes cycle $x \to y \to z \to x$, and the rotation about y carries z toward x rather than x toward z.

5 Building Complex Orientations: Elemental Matrices vs. Euler Angles

– **Elemental matrices** (R_x, R_y, R_z) are the individual building blocks, each handling rotation around a single axis.

– **Euler angles** (ϕ, θ, ψ) are the angles fed into a sequence of elemental rotations to reach a complex 3D orientation. Each angle is the amount of rotation about one axis in the sequence.

5.1 Chaining Rotations

Complex orientations are built by multiplying elemental matrices in sequence. For example, the aerospace Z-Y-X sequence is

$$R_{\text{total}} = R_z(\psi)\, R_y(\theta)\, R_x(\phi)$$

Read the product from right to left. When this matrix multiplies a vector v, the rightmost factor acts first:

$$R_{\text{total}}\, v = R_z(\psi)\big(R_y(\theta)\big(R_x(\phi)\, v\big)\big)$$

So the vector is first rotated by ϕ about x, then by θ about y, then by ψ about z.

5.2 What ϕ, θ, and ψ Represent

In the aerospace Z-Y-X convention, used for aircraft, ships, and spacecraft, the three angles have names:

– ϕ **(roll):** rotation about the x-axis, the body's first rotation. It tilts the wings left or right.

– θ **(pitch):** rotation about the y-axis, the second rotation. It tilts the nose up or down.

– ψ **(yaw):** rotation about the z-axis, the last rotation. It turns the heading left or right, like a compass direction.

The order matters because rotations do not commute. Rolling and then pitching gives a different orientation from pitching and then rolling. Changing the order of the matrices changes the final orientation.

5.3 The 3-1-3 Sequence

Another common choice is the Z-X-Z sequence, written 3-1-3 because it uses axes 3, 1, 3. It is standard in astronomy and orbital mechanics:

$$R_{\text{total}} = R_z(\psi)\, R_x(\theta)\, R_z(\phi)$$

Here the first rotation ϕ is about the z-axis (precession of the orbit's node), θ is about the x-axis (tilt, or inclination), and ψ is a second rotation about z (spin or argument of periapsis within the orbit). Its meaning depends on the field, so check which convention a source uses before combining formulas.

5.4 Determinant of a Chain

Since $\det(AB) = \det(A)\det(B)$, each elemental factor has determinant 1, so

$$\det(R_{\text{total}}) = \det(R_z)\det(R_y)\det(R_x) = 1$$

A chain of rotations is still a rotation.

5.5 Changing Frames

Applying these matrices to coordinate vectors lets a system transform coordinates and orientation descriptions cleanly between reference frames.